

Backtesting

Fetch historical candle data and instrument information for backtesting trading strategies using Groww APIs

"Note: Currently, Backtesting APIs only support CASH and FNO segments."

Groww Symbol

Groww symbol is a easy to construct unique identifier for an instrument across exchanges and segments. It is formed by concatenating

- **Exchange** - Where the instrument is traded
- **Trading Symbol** - The name/ticker of the instrument
- **Expiry Date** - Only for derivatives (format: DDMMmYY, example: 23Jan25)
- **Strike Price** - Only for options (the target price level)
- **Option Type** - Only for derivatives:
 - CE = Call Option
 - PE = Put Option
 - FUT = Futures

For Stocks and Indices: Only exchange and trading symbol are used.

For Futures: Exchange, trading symbol, expiry date, and "FUT" are used.

For Options: All components are used including strike price and option type (CE/PE).

For example:

- Equity: NSE-WIPRO , BSE-RELIANCE
- Index: NSE-NIFTY , BSE-SENSEX
- Future: NSE-NIFTY-30Sep25-FUT , BSE-SENSEX-25Sep25-FUT
- Call Option: NSE-NIFTY-30Sep25-24650-CE , BSE-SENSEX-25Sep25-79500-CE
- Put Option: NSE-NIFTY-30Sep25-24650-PE , BSE-SENSEX-25Sep25-79500-PE

Groww symbol also exists in the instruments csv file and it can be obtained from the [Get Instruments](#) API.

Get Expiries

This API retrieves available expiry dates for derivatives instruments (FNO) for a given exchange and underlying symbol. Useful for backtesting options and futures strategies. Data of FNO instruments are available from 2020.

Python SDK Usage

```
from growwapl import GrowwAPI

# Groww API Credentials (Replace with your actual credentials)
API_AUTH_TOKEN = "your_token"

# Initialize Groww API
groww = GrowwAPI(API_AUTH_TOKEN)

expiries_response = groww.get_expiries(
    exchange=groww.EXCHANGE_NSE,
    underlying_symbol="NIFTY",
    year=2024,
    month=1
)

print(expiries_response)
```

Request Schema

Name	Type	Description
exchange *	string	Stock exchange (NSE, BSE)
underlying_symbol *	string	Underlying symbol for which expiry dates are required (e.g., NIFTY, BANKNIFTY, RELIANCE)
year	integer	Year for which expiry dates are required (2020 - current year). If year is not specified, current year is considered.
month	integer	Month for which expiry dates are required (1-12). If month is not specified, expiries of the entire year is returned.

* required parameters

Response Payload

```
{
  "expiries" : [
    "2024-01-25",
    "2024-01-31",
    "2024-02-29",
    "2024-03-28"
  ]
}
```

Response Schema

Name	Type	Description
expiries	array[string]	Array of expiry dates in YYYY-MM-DD format

Get Contracts

This API retrieves available contract symbols for derivatives instruments for a given exchange, underlying symbol, and expiry date. Essential for backtesting specific options or futures contracts by passing them in [Candles API](#). Data of FNO instruments are available from 2020.

Python SDK Usage

```
from growwapl import GrowwAPI

# Groww API Credentials (Replace with your actual credentials)
API_AUTH_TOKEN = "your_token"
```



```
  ],
  [
    "2025-09-24T11:30:00",
    244.95,
    245.28,
    244.6,
    245.13,
    353800,
    null
  ],
  1.
}
```

Response Schema

Name	Type	Description
candles	array(array)	Array of candle data. Each candle contains: timestamp (yyyy-MM-dd HH:mm:ss), open, high, low, close, volume, open interest
closing_price	float	Closing price of the instrument
start_time	string	Start time in yyyy-MM-dd HH:mm:ss format
end_time	string	End time in yyyy-MM-dd HH:mm:ss format
interval_in_minutes	int	Interval in minutes

Backtesting Data Limits

Candle Intervals	Max Duration per Request
1 min, 2 min, 3 min, 5 min	30 days
10 min, 15 min, 30 min	90 days
1 hour, 4 hours, 1 day, 1 week, 1 month	180 days

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